HEPARIN

High-alert drug

Check units & concentration

CHEMICAL NAME Heterogeneous sulfated glycosaminoglycan (unfractionated heparin sodium, MW 3,000–30,000 Da) from porcine intestinal mucosa

CLASS Unfractionated heparin (UFH) anticoagulant

TRADE NAMES Heparin sodium injection · Hep-Lock · Monoparin · Liquemine

THERAPEUTIC USES / INDICATIONS

- Treatment of DVT/PE and acute coronary syndrome (IV infusion).
- VTE prophylaxis (SC) — useful in severe renal impairment.
- Anticoagulation during cardiopulmonary bypass, haemodialysis, PCI, ECMO.
- When rapid reversal may be needed (short t½, protamine fully reverses).

ADMINISTRATION

- **IV infusion** by syringe pump according to weight-based nomogram; adjust by aPTT/anti-Xa.
- **SC** into abdomen (not IM — haematoma).
- Double-check concentration (e.g. 1,000 vs 5,000 units/mL or higher strengths) and units with second nurse.
- Do not rub injection site.

DOSAGES

- **VTE treatment:** 5,000 units IV loading dose, then continuous infusion (US label 20,000–40,000 units/24 h; many units use a weight-based nomogram); adjust to aPTT about 1.5–2.5× control.
- **Children:** 75–100 units/kg IV bolus over 10 min, then 25–30 units/kg/h (infants) or 18–20 units/kg/h (>1 yr). **ACS/PCI:** per cardiology protocol.
- **Prophylaxis:** 5,000 units SC every 8–12 h.
- Check aPTT 6 h after start and after each change.

PHARMACOKINETICS

- **Onset:** IV immediate; SC 20–60 min. **Duration:** IV 2–6 h.
- Not absorbed orally; binds plasma proteins and endothelium.
- Cleared by reticuloendothelial system (dose-dependent).
- **t½:** ~1–1.5 h.

PHARMACODYNAMICS

- Binds **antithrombin III** → greatly accelerates inactivation of **thrombin (IIa) and factor Xa** (and IXa, XIa, XIIa).
- Prevents clot extension; does not dissolve existing clots.
- Monitored by aPTT.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Bruising, haematoma at injection site.
- Minor bleeding.
- Mild transient thrombocytopenia.

ADVERSE EFFECTS (SERIOUS)

- Major haemorrhage.
- Heparin-induced thrombocytopenia (HIT): platelet fall >50% at day 5–14 with paradoxical thrombosis; stop all heparin.
- Hyperkalaemia (aldosterone suppression).
- Osteoporosis (long-term); anaphylaxis.

CONTRAINDICATIONS & CAUTIONS

- Active major bleeding; haemophilia; severe thrombocytopenia.
- History of HIT.
- Recent intracranial haemorrhage; severe uncontrolled hypertension.
- Recent surgery of brain, spinal cord or eye; epidural catheter (timing rules).

KEY INTERACTIONS

- **Antiplatelets, NSAIDs, thrombolytics, warfarin, DOACs:** ↑ bleeding.
- **IV glyceryl trinitrate:** may ↓ heparin effect.
- **ACE inhibitors, K^+-sparing diuretics:** hyperkalaemia.

NURSING MANAGEMENT

- Weigh patient; use nomogram; independent double check of dose, pump rate and concentration.
- aPTT/anti-Xa as scheduled; adjust promptly.
- Platelets at baseline and every 2–3 days (HIT).
- Monitor for bleeding (wounds, urine, stools, neuro status).
- Keep protamine available.

PATIENT EDUCATION

- Report bleeding gums, nosebleeds, blood in urine or stools, sudden headache.
- Use a soft toothbrush and electric razor.
- Avoid NSAIDs.

KEY POINTS

- Antithrombin-dependent IIa + Xa inhibitor.
- Short t½ — useful in renal failure and before procedures.
- Monitor aPTT; reversed fully by protamine.
- Watch platelets for HIT.

SOURCES CHECKED **[412]** Heparin Sodium, IV, SC (US label) **[413]** Heparin Sodium, IV, SC (US label, REMEDYREPACK INC)

Full references p. 432–457